

Method	Online models during RL	Offline prefix cost	Rollouts/problem	Time/step	Relative continuation tokens (last 100 steps)	Final average score
GRPO	Policy only	0	16	43.8 s	1.00x	39.6
DenoiseRL-GRPO	Policy only	0.5 h on 4 GPUs (2 GPU-hours), one-time	12 + 4	49.7 s	1.27x	42.0